

Halo Mass Mediates the Colour–Environment Relation in GAMA: Evidence from Partial Correlation and Distribution Shape Analysis

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Abstract

We investigate the relationship between galaxy colour ($u-r$) and local environment for galaxies in the GAMA DR4 survey, using partial correlation to control for group halo mass and bootstrap null tests to assess physical evolution in the shape of the colour distribution with redshift. For elliptical galaxies ($N = 1,270$), raw correlations between $u-r$ and all four environment metrics collapse to $r \approx 0$ once group halo mass is controlled, demonstrating complete mediation. For the full galaxy population ($N = 22,531$), $\sim 50\%$ of the signal survives halo control, indicating an independent morphology-linked pathway. Skewness and Box–Cox λ of the $u-r$ distribution evolve monotonically with redshift (Spearman $|r| = 1.00$, $p < 0.001$ for both): skewness shifts from $+0.90$ at $z = 0.03$ to -0.82 at $z = 0.33$, and λ from -0.20 to $+2.70$. Both trends exceed the bootstrap 95% null envelope at every redshift bin, confirming physical evolution consistent with ongoing galaxy quenching. Comparison of λ with the red fraction ($u-r > 2.1$) reveals two decoupled processes: active threshold crossing, which decelerates above $z \approx 0.15$, and red sequence consolidation, which continues at higher redshift and is detected by λ but not by the red fraction alone.

1. Introduction

The correlation between galaxy colour and local environment is among the most robust observational results in galaxy evolution (Dressler 1980; Blanton & Moustakas 2009). The outstanding question is mechanistic: does environment directly drive quenching, or is the signal primarily a reflection of halo mass, which correlates with both environment and the efficiency of quenching processes such as ram-pressure stripping and strangulation?

A complementary question is whether the shape of the colour distribution encodes evolutionary information beyond the mean. If the $u-r$ distribution transforms systematically with redshift in ways exceeding sampling noise, that transformation constrains the quenching timescale independently of correlation analyses. In particular, the relationship between Box–Cox λ and the conventional red fraction (fraction of galaxies above a fixed colour threshold) can distinguish two qualitatively different evolutionary processes: active quenching — galaxies crossing the blue-red threshold — and red sequence consolidation — the red sequence becoming more concentrated and sharply bounded even after the crossing rate decelerates.

We address both questions using GAMA DR4 (Driver et al. 2022), exploiting its group catalogue (Robotham et al. 2011) and environment measures (Brough et al. 2013).

2. Data and Sample

GAMA DR4 StellarMassesLamdarv24 (200,077 galaxies), quality cuts $0.5 < u-r < 4.0$, $\log M_* > 8.0$, $0.002 < z < 0.35$: 165,488 galaxies for shape analysis. Partial correlation additionally uses G3CGalv10 (group membership), G3CFoFGroupv10 (halo mass proxy MassA), and EnvironmentMeasuresv06 (CountInCyl, DistanceTo5nn, SurfaceDensity, AGEDenPar). Elliptical subsample: VisualMorphologyv03 ELLIPTICAL_CODE = 1, $N = 1,270$. Full group population $N = 22,531$.

3. Methods

3.1 Partial Correlation

We residualise both $u-r$ and each log-environment measure against $\log(\text{MassA})$ via linear regression and correlate residuals (Pearson and Spearman). Results at individual galaxy level and group level (mean $u-r$; red fraction f_{red} = fraction with $u-r > 2.1$).

3.2 Distribution Shape Evolution

Seven equal-width redshift bins ($z = 0.002-0.35$, $\Delta z = 0.05$; $N = 3,900-34,000$). In each bin we compute skewness, excess kurtosis, and Box-Cox λ . Johnson SU parameters are reported in Appendix A; they show numerical instability and are excluded from the main analysis. We bootstrap 500 same-sized subsamples from the full pool to build 95% null envelopes and apply Spearman and Kendall rank tests for monotonic trend.

3.3 Red Fraction and λ Comparison

Red fraction f_{red} is computed in each redshift bin as the fraction of galaxies with $u-r > 2.1$. Both f_{red} and λ are normalised to $[0,1]$ over the seven bins to place them on a common scale, enabling direct comparison of their rates of change. The $d\lambda/dz$ to $d(f_{\text{red}})/dz$ ratio quantifies the relative sensitivity of each statistic to the ongoing quenching process.

4. Results

4.1 Halo Mass Fully Mediates Colour–Environment for Ellipticals

Table 1 shows raw and partial Pearson correlations for the elliptical subsample. Raw $r = 0.07-0.09$; after controlling for $\log(\text{MassA})$ all partial correlations collapse to $r \approx 0$ ($p > 0.3$). At the group level ($N = 242$ groups) raw r rises to $0.13-0.15$ but again collapses after halo control.

Table 1. Partial correlation — ellipticals, controlling for $\log(\text{MassA})$

Environment measure	Raw r	Raw p	Partial r	Partial p	Sig?
Individual level ($N = 1,270$)					
$\log(\text{CountInCyl})$	+0.066	0.019	-0.029	0.296	No
$\log(\text{DistanceTo5nn})$	-0.090	0.001	-0.006	0.824	No
$\log(\text{SurfaceDensity})$	+0.089	0.002	+0.005	0.869	No
$\log(\text{AGEDenPar})$	+0.071	0.012	+0.023	0.424	No
Group level ($N = 242$ groups)					
$\log(\text{CountInCyl})$	+0.133	0.039	+0.025	0.700	No
$\log(\text{DistanceTo5nn})$	-0.147	0.022	-0.046	0.480	No
$\log(\text{SurfaceDensity})$	+0.147	0.023	+0.044	0.492	No
$\log(\text{AGEDenPar})$	+0.037	0.572	-0.016	0.808	No

All partial correlations statistically indistinguishable from zero ($p > 0.3$).

4.2 Independent Environment Signal for the Full Population

For all galaxy types ($N = 22,531$ individuals; 5,775 groups), raw $r = 0.18$ – 0.25 , remaining significant after halo control (partial $r = 0.10$ – 0.16 , all $p < 10^{-55}$). Approximately 45–58% of the raw signal survives, indicating a secondary pathway.

Table 2. Partial correlation — full population, controlling for $\log(\text{MassA})$

Environment measure	Raw r	Raw p	Partial r	Partial p	% ret.
Individual level ($N = 22,531$)					
$\log(\text{CountInCyl})$	+0.240	8×10^{-292}	+0.109	7×10^{-61}	45.6%
$\log(\text{DistanceTo5nn})$	-0.245	3×10^{-304}	-0.124	4×10^{-78}	50.8%
$\log(\text{SurfaceDensity})$	+0.245	1×10^{-304}	+0.124	3×10^{-78}	50.8%
$\log(\text{AGEDenPa})$	+0.178	2×10^{-160}	+0.104	2×10^{-55}	58.4%
Group level ($N = 5,775$) — mean $u-r$					
$\log(\text{CountInCyl})$	+0.234	$< 10^{-300}$	+0.160	$< 10^{-300}$	68.4%
$\log(\text{DistanceTo5nn})$	-0.228	$< 10^{-300}$	-0.153	$< 10^{-300}$	67.1%
$\log(\text{SurfaceDensity})$	+0.228	$< 10^{-300}$	+0.153	$< 10^{-300}$	67.1%
$\log(\text{AGEDenPa})$	+0.154	$< 10^{-300}$	+0.128	$< 10^{-300}$	83.1%

All partial correlations remain highly significant ($p < 0.05$).

4.3 Colour Distribution Shape Evolves Monotonically with Redshift

Table 3 and Figure 1 show Box–Cox λ and skewness against redshift with bootstrap null envelopes. Both parameters lie outside the null at every bin. Skewness shifts from +0.90 at $z = 0.026$ to -0.82 at $z = 0.325$ (Spearman $r = -1.000$, $p = 4.0 \times 10^{-4}$). Box–Cox λ increases from -0.20 to $+2.70$ (Spearman $r = +1.000$, $p = 4.0 \times 10^{-4}$). The sign flip in skewness reflects the progressive consolidation of the red sequence. The transition through $\lambda = 0$ near $z \approx 0.08$ marks the approximate epoch at which the colour distribution crosses from sub-log-Normal to supra-Normal breadth.

Table 3. $u-r$ distribution shape parameters by redshift bin

z centre	N	Box-Cox λ	Skewness	Excess kurtosis
0.026	3,932	-0.200	+0.899	+0.824
0.075	16,579	+0.431	+0.263	-0.750
0.125	29,589	+1.125	-0.148	-1.046
0.175	32,999	+1.694	-0.387	-0.897
0.225	27,116	+1.995	-0.523	-0.623
0.275	33,685	+2.532	-0.745	-0.188
0.325	21,588	+2.701	-0.821	+0.069

Spearman $r(\lambda, z) = +1.000$, $p = 4.0 \times 10^{-4}$; Spearman $r(\text{Skew}, z) = -1.000$, $p = 4.0 \times 10^{-4}$. All bins lie outside the bootstrap 95% null envelope.

Figure 1. Box–Cox λ (left) and skewness (right) of the $u-r$ colour distribution vs redshift. Grey band = 95% bootstrap null envelope (500 iterations). Red stars mark bins where the observed value lies outside the null. Both parameters show perfect monotonic trends (Spearman $|r| = 1.00$, $p = 4 \times 10^{-4}$).

4.4 Red Fraction and λ Reveal Two Decoupled Quenching Processes

Table 4 and Figure 2 compare the red fraction f_{red} with Box–Cox λ across the seven redshift bins. Spearman $r(f_{\text{red}}, \lambda) = 1.000$, confirming perfect rank agreement. Despite this monotonic correlation, the two statistics evolve at markedly different rates.

Table 4. Red fraction and Box–Cox λ by redshift bin

z centre	N	Red fraction	Box-Cox λ	$ \lambda - 1 $
0.026	3,932	0.142	-0.200	1.200
0.075	16,579	0.219	+0.431	0.569
0.125	29,589	0.336	+1.125	0.125
0.175	32,999	0.405	+1.694	0.694
0.225	27,116	0.434	+1.995	0.995
0.275	33,685	0.468	+2.532	1.532
0.325	21,588	0.477	+2.701	1.701

Spearman $r(f_{\text{red}}, \lambda) = +1.000$. Red fraction = fraction of galaxies with $u-r > 2.1$. $|\lambda - 1|$ measures distance from a Normal distribution ($\lambda = 1$ is the Normal point).

When both statistics are normalised to $[0,1]$, the red fraction consistently leads λ between $z = 0.075$ and $z = 0.275$ (Figure 2, centre panel), with maximum divergence near $z = 0.175$ (normalised difference = 0.13). This means that, relative to its total range, the red fraction rises faster than λ across low-to-mid redshift, and λ catches up only at the extremes.

The $d\lambda/dz : d(f_{\text{red}})/dz$ ratio, which measures the relative sensitivity of each statistic per unit redshift, increases monotonically from ~ 6 at $z \approx 0.10$ to ~ 19 at $z \approx 0.30$. As the red fraction growth decelerates — dropping from $\Delta f_{\text{red}} = 0.12$ per bin in the $z = 0.10$ – 0.15 interval to $\Delta f_{\text{red}} = 0.009$ in the $z = 0.30$ – 0.35 interval — λ continues to rise steeply. At the highest redshift bins, λ detects structural change in the colour distribution that threshold counting is nearly insensitive to.

We interpret this divergence as the signature of two physically distinct processes acting on different timescales. Active quenching — galaxies transitioning from blue cloud to red sequence and crossing the $u-r > 2.1$ threshold — drives f_{red} and dominates at low-to-mid redshift ($z < 0.15$). Red sequence consolidation — the galaxies already on the red sequence becoming more concentrated and the distribution acquiring a sharper left boundary — drives λ without necessarily changing f_{red} , and dominates at higher redshift in the GAMA sample. The $|\lambda - 1|$ column in Table 4 quantifies the non-Normality of the distribution: it peaks at the lowest and highest redshifts, with a minimum of 0.125 near $z = 0.125$, the bin most consistent with a Gaussian colour distribution.

Figure 2. Left: Box–Cox λ and red fraction ($u-r > 2.1$) vs redshift on dual y-axes. Centre: both statistics normalised to $[0,1]$ — the shaded gap shows where red fraction leads (pink) or λ leads (blue). Right: rates of change $d\lambda/dz$ and $d(f_{\text{red}})/dz$ per redshift bin, showing the decoupling between threshold crossing and distribution consolidation at $z > 0.15$.

5. Discussion

Our partial correlation results align with the picture that halo mass is the primary driver of colour–environment correlations for morphologically selected red galaxies (Peng et al. 2012; Woo et al. 2013). Complete mediation for ellipticals suggests that once a galaxy has settled into a massive halo and undergone morphological transformation, residual local environment adds little further quenching pressure. The ~50% survival of the signal in the full population points to secondary processes consistent with harassment and tidal stripping acting preferentially on late-type discs (Moore et al. 1996).

The distribution shape results offer a model-independent complement. Perfect monotonic Spearman trends ($|r| = 1.00$, $p < 0.001$) across seven independent redshift bins, combined with consistent departure from the bootstrap null, constitute strong evidence of real physical evolution — a direct quantitative answer to the objection that shape differences merely reflect statistical variation.

The comparison of λ with the red fraction (§4.4) introduces a further diagnostic dimension. Because λ captures the full distributional shape rather than counting galaxies above a fixed threshold, it remains sensitive to structural changes in the red sequence even when active quenching — and therefore f_{red} — has largely saturated. The increasing $d\lambda/d(f_{\text{red}})$ ratio with redshift (from ~8 to ~19 across the GAMA range) implies that at $z > 0.2$, the dominant evolutionary process is consolidation of an already-red population rather than recruitment of new red galaxies. This is consistent with the passive evolution of massive red-sequence galaxies dominating at low-to-moderate redshift (Blanton & Moustakas 2009), and with the Malmquist-selected nature of the high- z GAMA bins. We note, however, that disentangling genuine evolutionary consolidation from flux-limit selection bias requires a stellar-mass-complete subsample — a test we defer to future work.

6. Conclusions

1. Halo mass fully mediates colour–environment for elliptical galaxies ($N = 1,270$); partial correlations collapse to $r \approx 0$ after controlling for $\log(\text{MassA})$.
2. For the full galaxy population ($N = 22,531$), ~50% of the signal survives halo control, indicating a morphology-linked secondary pathway.
3. Skewness and Box–Cox λ evolve monotonically with redshift ($|\text{Spearman } r| = 1.00$, $p = 4 \times 10^{-4}$): skewness from +0.90 to –0.82 and λ from –0.20 to +2.70, inconsistent with sampling noise and consistent with progressive consolidation of the red sequence.
4. Comparison of λ with the red fraction reveals two decoupled processes: active threshold crossing dominates at $z < 0.15$; red sequence consolidation, detected by λ but not f_{red} , dominates at higher redshift. The $d\lambda/d(f_{\text{red}})$ sensitivity ratio increases monotonically from ~8 to ~19 across $z = 0.05$ – 0.30 .
5. Box–Cox λ provides information about colour distribution shape beyond what a fixed-threshold red fraction can capture, and is recommended as a complementary statistic in redshift-resolved quenching analyses.

Data Availability

GAMA DR4: <https://www.gama-survey.org/dr4/>. Analysis notebooks available on request.

Appendix A: Johnson SU Fit Parameters

Table A1 gives the full set of shape parameters including Johnson SU (a , b). The JSU a parameter shows large non-monotonic variation despite the smooth evolution of skewness and λ , arising from correlation between a and b on a near-flat likelihood ridge.

Table A1. Full shape parameters including Johnson SU

z	N	JSU a	JSU b	Box-Cox λ	Skewness	Kurt
0.026	3,932	-2.763	+2.210	-0.200	+0.899	+0.824
0.075	16,579	-11.677	+5.925	+0.431	+0.263	-0.750
0.125	29,589	+20.679	+7.171	+1.125	-0.148	-1.046
0.175	32,999	+9.776	+3.566	+1.694	-0.387	-0.897
0.225	27,116	+3.431	+2.112	+1.995	-0.523	-0.623
0.275	33,685	+2.802	+1.752	+2.532	-0.745	-0.188
0.325	21,588	+2.836	+1.802	+2.701	-0.821	+0.069

JSU a: Spearman $r = +0.29$, $p = 0.53$ (not significant). Box–Cox λ and skewness: $|r| = 1.00$, $p < 0.001$. JSU a varies by $>30\times$ while skewness changes by only 0.75 — instability is numerical, not physical.

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